

Waaberi Ibrahim

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EDUCATION

University of Ottawa

BASc in Software Engineering, Co-op

Ottawa, ON, Canada

GPA: 3.9/4.0 | Expected Graduation: Dec 2029

EXPERIENCE

Software Engineering Co-op/Intern

May 2026 – Aug. 2026

Statistics Canada

Ottawa, ON

- Reached **hundreds of employees** with an award-winning **Gatsby/React** website for an introductory AI course
- Automated chart styling with an **R** library that applies Statistics Canada publication standards to any visualization
- Saved **hundreds of hours** with a **Python** tool that converts legacy e-learning courses into editable web lessons
- Enabled employees to resolve training questions using a **RAG-powered AI assistant** grounded in approved documentation

Undergraduate Research Assistant, Health Devices

May 2026 – Present

CARG Health Devices, University of Ottawa

Ottawa, ON

- Built a contactless monitoring platform that turns vital-sign and fall data into real-time alerts
- Integrated live sensors over **MQTT**, turning raw device feeds into structured real-time data
- Engineered **role-based** response workflows with protocol checks, medical sign-off, and audit logging
- Automated zero-touch deployment of the **Docker** stack via a **GitOps** poller that ships on every commit

Webmaster

Jan. 2026 – Present

IEEE uOttawa Student Branch

Ottawa, ON

- Own and deploy a live **Next.js/Payload CMS** site serving **4,000+ visits** and **20,000+ impressions** monthly
- Contributed **23 merged PRs** and **73 commits** as top contributor in an **8-contributor** open-source repository
- Shipped bilingual routing across **20+ CMS fields**, static generation, sitemaps, meta tags, and automated backups
- Reached **50 participants** by delivering a web security workshop on common vulnerabilities in CTFs

PROJECTS

Sentinel | *FastAPI, React, XGBoost, Docker* | [Demo](#) | [GitHub](#)

- Built a containerized **FastAPI** backend that ingests CSV/XLSX trade histories and serves session-scoped analysis
- Implemented **8 REST endpoints** for metrics, bias scoring, P/L analysis, recommendations, and what-if simulation
- Trained an **XGBoost** model for bias scoring, recommendations, and what-if simulation with **91% accuracy**
- Published a working **YouTube demo**; delivered with a **4-person team** and won **3rd Place** at the hackathon

Quark | *QNX, C++, TensorFlow Lite, Raspberry Pi* | [Devpost](#) | [GitHub](#)

- Built a **QNX driver-safety system** that detects drowsiness from camera input and triggers audio alerts
- Ran vision and language inference fully on-device with **TensorFlow Lite** and **llama.cpp**, avoiding cloud dependencies
- Won **3rd Place in the QNX Challenge at cuHacking7** with a **3-person team** and working embedded-AI prototype

Synacor VM Emulator | *Python, Virtual Machines, Low-Level Systems* | [GitHub](#)

- Implemented a working **22-opcode**, 15-bit Synacor virtual machine in **Python**
- Built an 8-register CPU with **32,768-word memory**, stack operations, control flow, and terminal I/O
- Added an interactive **debugger** with stack inspection and runtime analysis to solve the challenge

SKILLS

Languages: Java, Python, TypeScript, R, C#, Go, Ruby, SQL, HTML, CSS, JS, C++

Frameworks & ML: React, Next.js, Node.js, FastAPI, Django, XGBoost, scikit-learn, Qiskit, Polars, Pandas, NumPy

Databases: PostgreSQL, MySQL, MongoDB, Redis, SQLite, Supabase

Tools: Git, Docker, Linux, Bash, Grafana, GitHub Actions, Vercel, Azure, AWS

Concepts: REST APIs, Machine Learning, Quantum Computing, Real-Time Systems, CI/CD, Cybersecurity, i18n

AWARDS AND CERTIFICATIONS

3rd Place - cuHacking7 QNX Challenge	2026
1st Place - IEEE uOttawa x HackTheBox Capture The Flag	2026
3rd Place - uOttawa SESA National Bank Hackathon	2026
1st Place - IBM Qiskit Fall Fest 2025 Hackathon (Quantum Computing)	2025
1st Place - uOttawa-IBM Cyber Range FlagForce Capture The Flag	2025
2nd Place - University of Ottawa Computer Chess Hackathon	2025